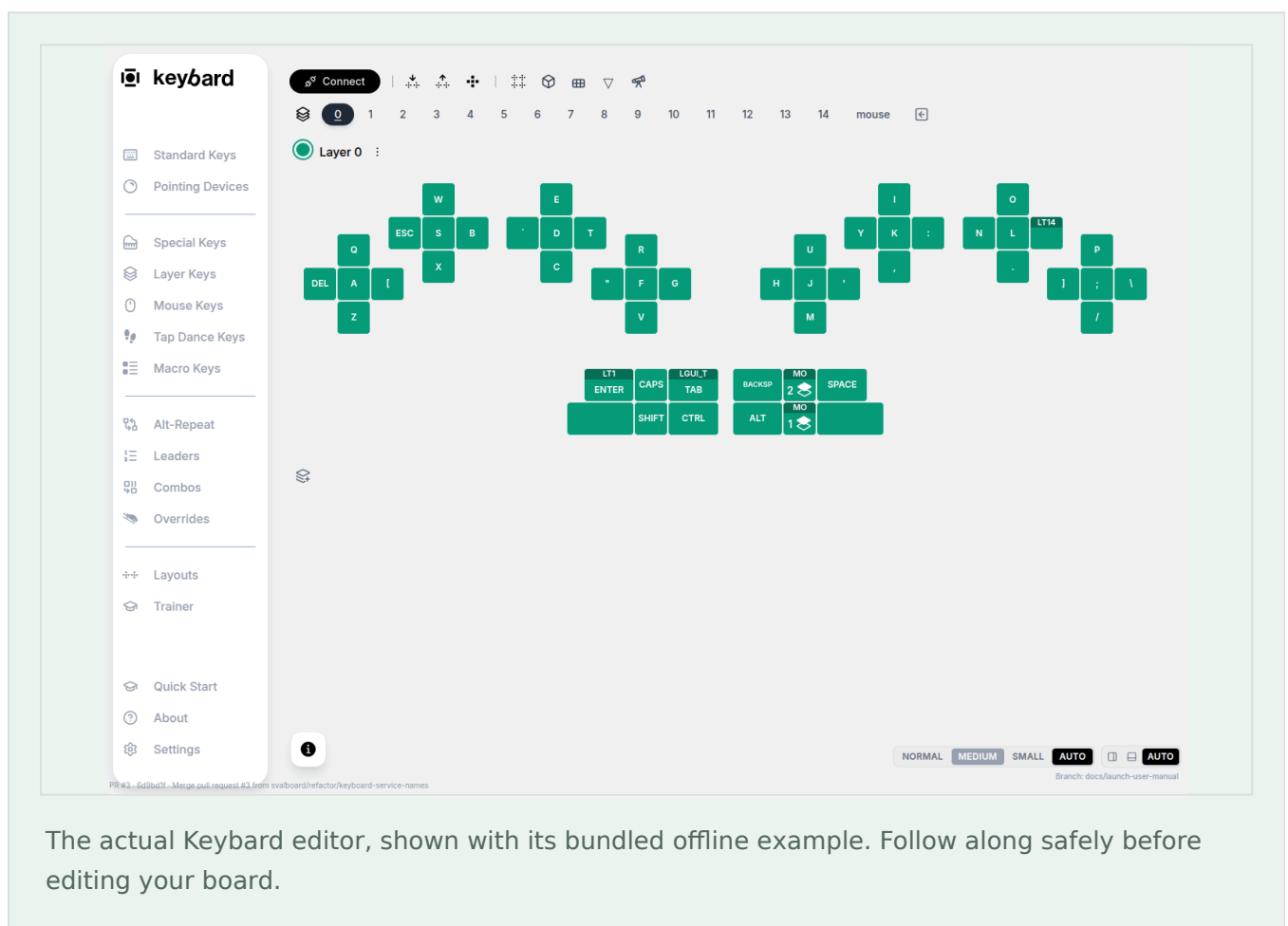


The Keyboard user manual.

Connect your board, make your first change, and learn every part of Keyboard.

11 chapters Illustrated walkthroughs Browser editor + desktop Trainer



The screenshot displays the Keyboard editor interface. On the left is a sidebar with a menu containing: keyboard (with a logo), Standard Keys, Pointing Devices, Special Keys, Layer Keys, Mouse Keys, Tap Dance Keys, Macro Keys, Alt-Repeat, Leaders, Combos, Overrides, Layouts, Trainer, Quick Start, About, and Settings. The main area shows a keyboard layout for 'Layer 0'. The layout includes keys for Q, W, E, R, T, Y, I, O, P, A, S, D, F, G, H, J, K, L, Z, X, C, V, U, M, N, B, and M. The bottom row features function keys: LT1, ENTER, CAPS, LGUI, TAB, BACKSP, MO 2, SPACE, SHIFT, CTRL, ALT, MO 1, and another function key. At the bottom right, there are buttons for NORMAL, MEDIUM, SMALL, and AUTO, along with a toggle for AUTO. A small information icon is visible in the bottom left corner of the main area.

The actual Keyboard editor, shown with its bundled offline example. Follow along safely before editing your board.

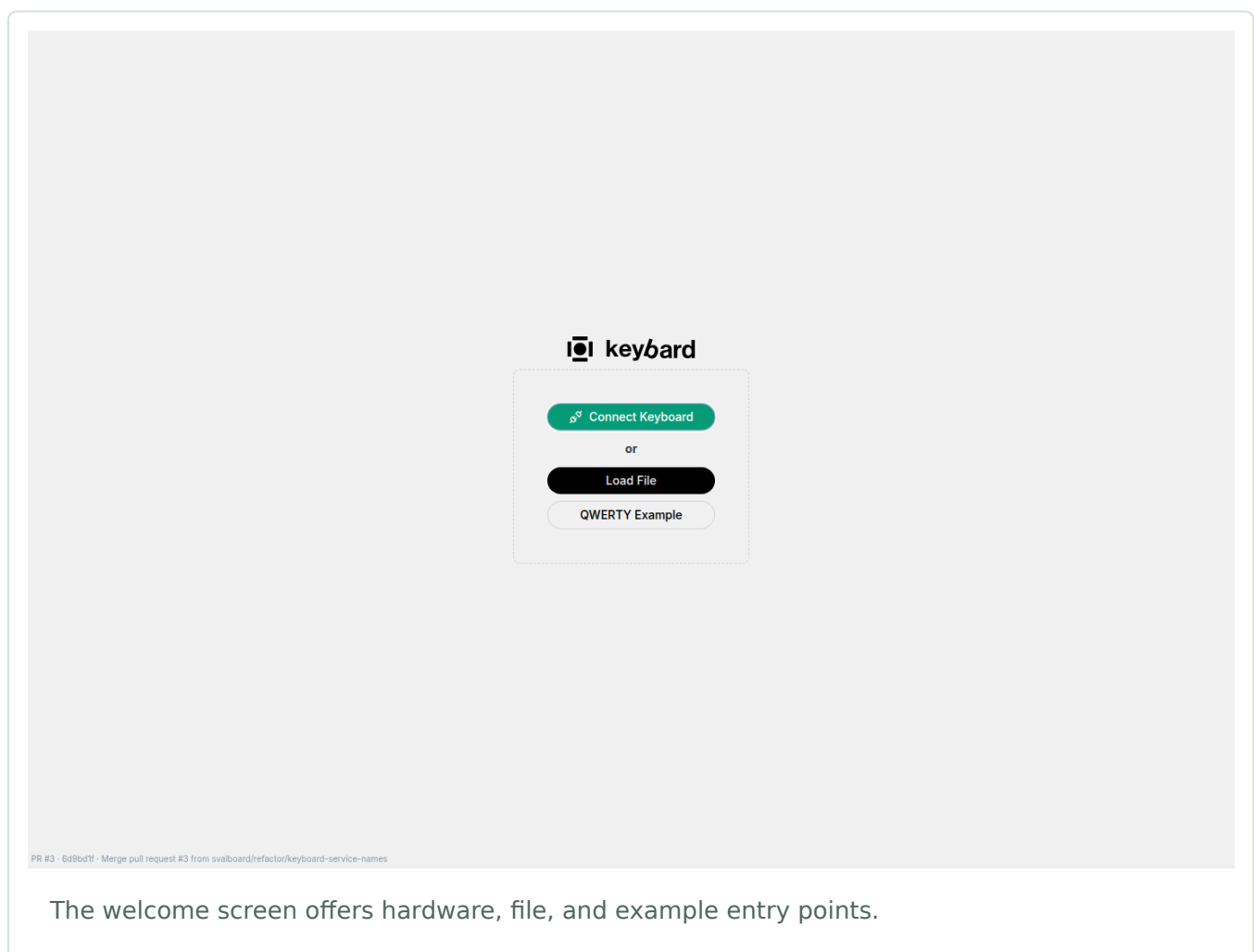
Written against launch Keyboard `6d9bd1f`. Screenshots use the bundled example and may show fewer behavior slots than current firmware. Connected hardware exposes additional controls. All screenshots can be enlarged.

Start with a keyboard—or an example

Keyboard lets you configure your Svalboard, organize layers, create advanced bindings, and set up an on-screen trainer. You can explore the editor before connecting hardware. This guide starts with one ordinary key change, then builds toward reusable layouts and reliable backups.

Choose a starting point on the welcome screen:

Choose	What happens	Use it for
Connect Keyboard	The browser asks which keyboard to open, then Keyboard reads its configuration.	Editing your connected Svalboard.
Load File	A saved layout opens as an offline draft.	Inspecting a backup or continuing exported work.
QWERTY Example	A bundled example opens without a keyboard.	Learning the controls safely.



Three places your work can live

ON YOUR SVALBOARD

The saved configuration

Runs on the keyboard.
Connecting reads it;
applying edits changes
it.

IN THIS BROWSER SESSION

The current draft

May include unsaved
changes. The layout you
see is not always what
the board runs.

ON YOUR COMPUTER

An exported file

A copy you keep. Export
downloads it; import
reviews which parts can
be applied.

Take the five-minute tour

1. Choose **QWERTY Example**.
2. If only icons are showing, click the logo at the top of the left navigation rail to reveal its labels. Hovering an icon also identifies it.
3. Open **Standard Keys**, select a key position in the layout, and choose a replacement from the key palette.
4. Switch layer tabs to see how the same physical positions can carry different bindings.
5. Choose **Export Layout** in the editor toolbar if you want to keep your experiment.

Offline means no keyboard writes. An example or loaded file is a draft in the current session. Export it before closing or refreshing. Connecting a keyboard later reads that keyboard; it does not automatically install your draft.

Most illustrations use the bundled offline example. Captions explicitly identify connected controls shown with controlled test-board data. No screenshot is presented as proof of a hardware write.

CHAPTER 02 / CONNECT YOUR BOARD

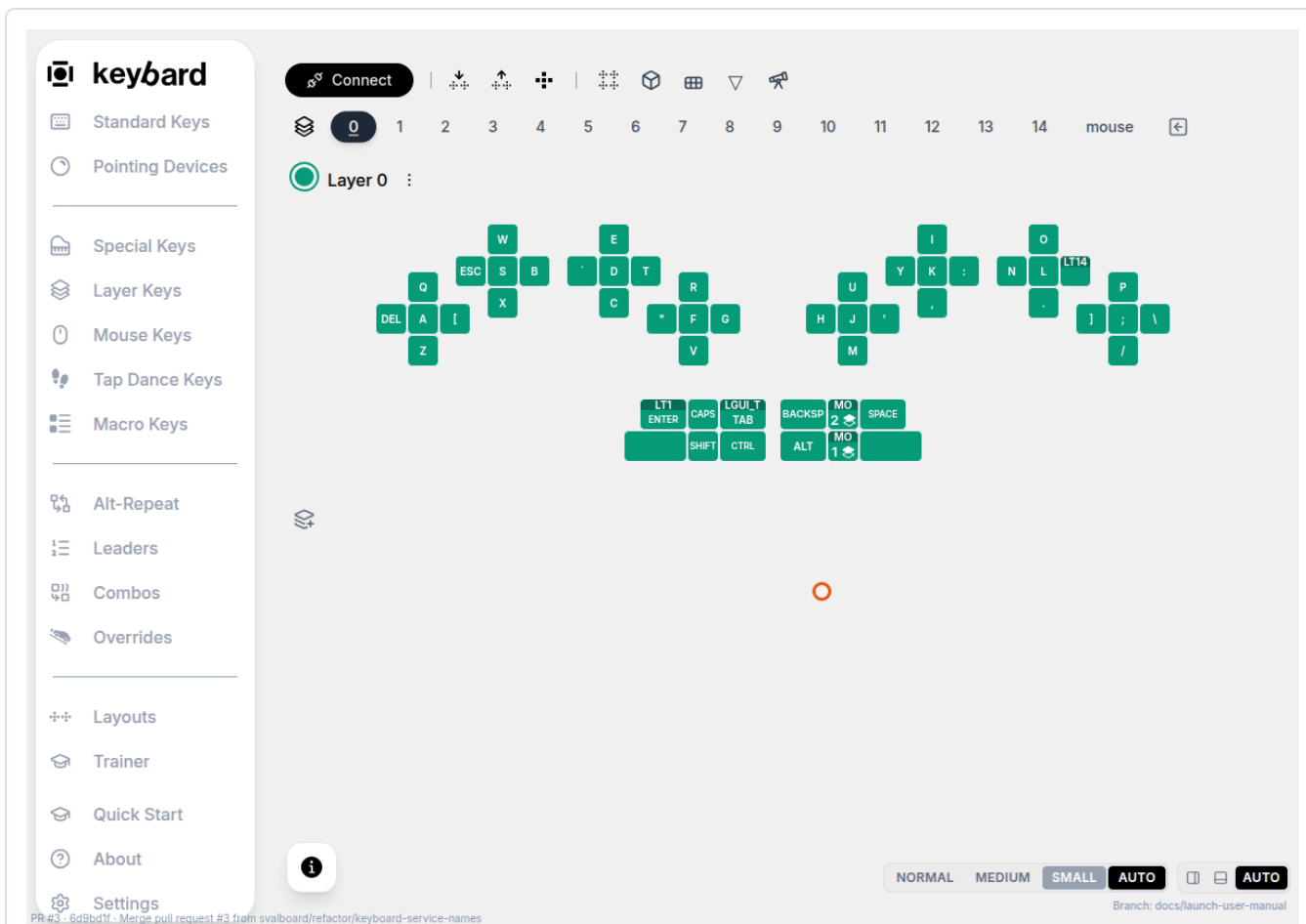
Connect and protect your starting point

Welcome → Connect Keyboard → browser device chooser

Use a desktop browser that supports WebHID, such as Chrome or Edge, and a USB data connection. The connection button is unavailable when the browser lacks the required API. Keybard's browser editor expects compatible Svalboard firmware; seeing a keyboard in a device chooser does not establish firmware compatibility.

1. Connect your Svalboard by USB, open Keybard, and choose **Connect Keyboard**.
2. Select the intended keyboard in the browser's chooser and grant access. With several keyboards attached, check which one you selected.
3. Wait for the layout to load. Hover the connection/update controls to check the editing target before changing anything.
4. Choose **Export Layout**. Keep the native **.svil (Recommended)** and **Include Macros** enabled, then choose **Export**.
5. Name the downloaded file clearly—for example, `before-first-edit.svil`—and keep it somewhere you can find again.

Back up before editing. Export captures the configuration currently shown in the editor, including staged changes. Exporting after experiments does not recover the untouched configuration you originally connected to.



Export a backup. Open Export Layout and export the native `.svil` file. The recording verifies a downloaded file; this example uses the browser download fallback. Browsers with a native save picker show a Save dialog instead. Offline example. The orange ring marks the pointer.

Coming from an older configuration tool

Keep your existing layout backup before changing firmware. You can open a legacy `.vil` file offline to inspect it, but Keybard blocks direct import of old-protocol layouts into connected Sval firmware when keycode migration is required. Renaming the file extension is not a conversion.

For Svalboard Vial-QMK `v2025-11-01` with the `vial` keymap, the firmware can migrate supported configuration on first boot. Follow the [firmware upgrade instructions](#), connect in Keybard, and verify layers, macros, modifier options, timing and pointing settings. Then export a fresh native `.svil`. Other old or custom builds are not covered; keep the old file as a reference and recreate anything not migrated. A supported firmware migration is different from importing a legacy file in the browser.

Understand the editing target

State	Meaning
Editing keyboard	Changes can be written to the connected device.
Offline draft	Changes affect the loaded file or example in this session. Export to keep them.
Pending changes	The editor has changes waiting to be written; the displayed layout may differ from the board.
Not saved	A write failed. Some writes may already have succeeded; follow the retry guidance.

CHAPTER 03 / YOUR FIRST EDIT

Make and verify your first change

keyboard

Standard Keys

Pointing Devices

Special Keys

Layer Keys

Mouse Keys

Tap Dance Keys

Macro Keys

Alt-Repeat

Leaders

Combos

Overrides

Layouts

Trainer

Quick Start

About

Settings

Connect

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 mouse

Layer 0

W E I O

ESC S B . D T R Y K : N L LT14

DEL A [X C " F G H J ' , P] ; \

Z V U M

LT1 ENTER CAPS LGUI TAB BACKSP MO SPACE

SHIFT CTRL ALT MO 1

NORMAL MEDIUM SMALL AUTO

Branch: docs/launch-user-manual

An offline example of the editing workspace: navigation on the left, layer selection above the layout.

Start with one easy-to-recognize binding on a known layer. Keep the keys you need for navigation and switching layers intact until you are comfortable restoring changes.

Pending (1)

⚡

Apply 1 Change

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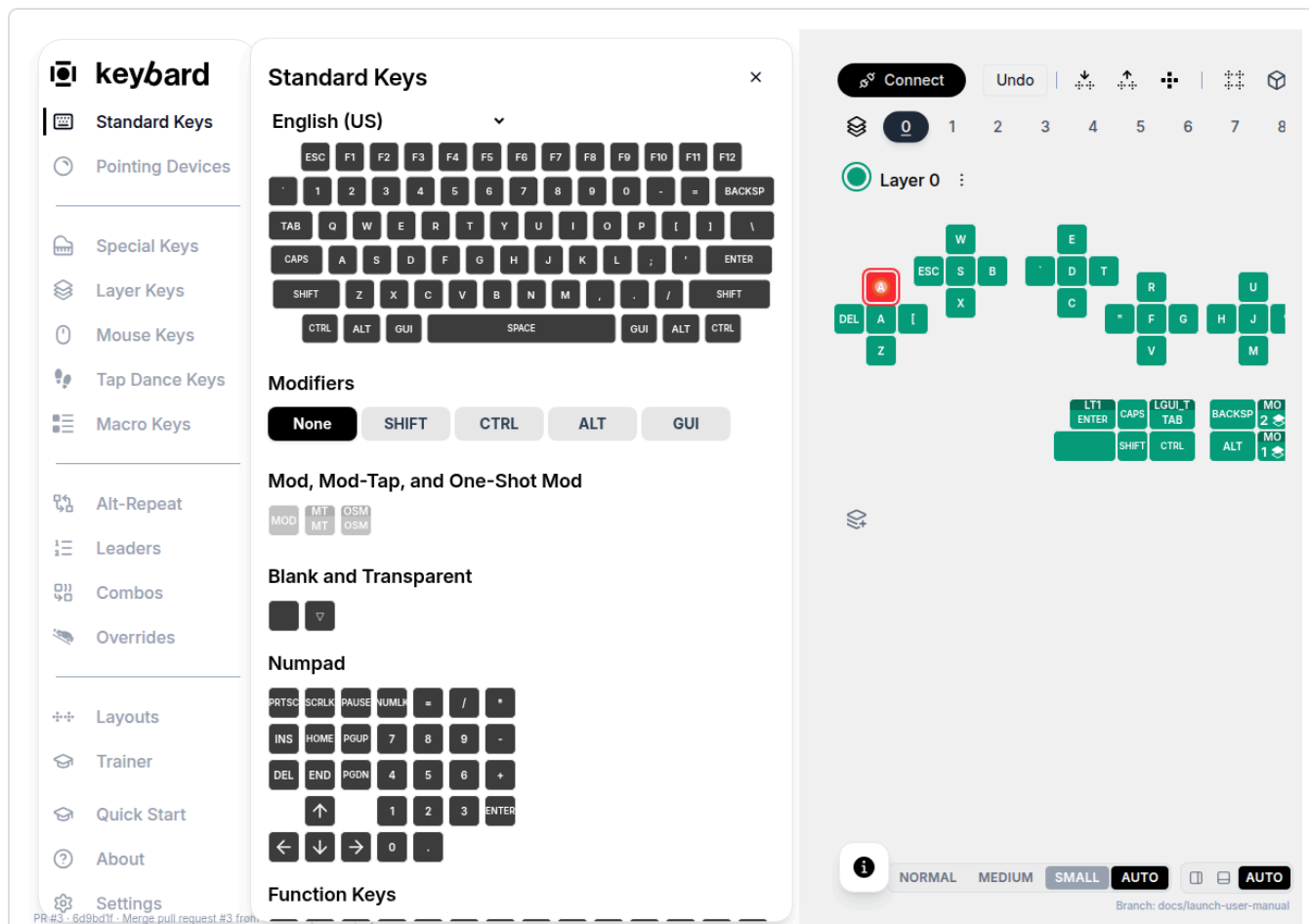
Actual toolbar with controlled test-board data. From the left: Pending, the update-mode lightning control, Apply, Undo, Import (down arrow), Export (up arrow), then Matrix Tester. No physical board was used.

Choose when changes reach the board

Live Updating is on by default. While connected, ordinary queued edits are saved as you make them. For a controlled first experiment, use the lightning control labeled **Switch to Manual Updates**. You can then review pending edits before applying them.

Control	Effect
Live Updating	Edits are saved as you make them; watch for saving or error messages.
Apply N Changes	Writes the pending changes in Manual mode.
Discard pending edits	Discards the current unsaved edits. It does not reverse changes already applied.
Switch to Live Updating	Applies pending changes before enabling Live Updating.

1. Select the layer you intend to edit.
2. Open **Standard Keys** and select a position on the displayed keyboard.
3. Choose its replacement from the palette. You can also drag a palette key onto the target.
4. Check the resulting legend. In connected Manual mode, open **Pending (N)** to review what will be written.
5. Choose **Apply N Changes** and wait for saving to finish. Do not assume a changed legend alone means the board was updated.
6. Deselect the key, switch to a text editor, activate the appropriate layer, and test the result.
7. Export the verified configuration as a new file.



Drag A onto the Q position. Watch the pointer pick up A from Standard Keys, drag it onto Q, and release. The position now shows A. This changes the offline example only. The orange ring marks the pointer.

Typing can assign a key. Typing Binds a Key is enabled by default. With an editing target selected, pressing a physical key can assign that key, including modifiers, to the target. Deselect before testing normal typing, or turn this option off in **Settings → General**.

Dragging an existing layout key onto another position swaps bindings, including across layers. Use this deliberately: it is different from copying a palette key. If an **Undo** button appears, its tooltip identifies the operation it can reverse; do not treat it as unlimited editing history.

Build a layout you can navigate

A layer is another set of bindings for the same physical key positions. Editing a layer tab changes what you are looking at in Keyboard; using a layer binding on the physical keyboard changes what the keyboard types.

Term	Meaning
Base/default layer	The starting layer used when no higher active layer provides a binding.
Transparent key	Looks through to a lower active layer for its binding.
Blank key	Produces no action at that position instead of falling through.
Momentary layer key	Activates another layer while held.
Layer-tap key	Types one action when tapped and activates a layer when held.

Open **Layer Keys** for layer bindings. Start with a momentary key leading to a small navigation or symbols layer. Keep the activation key usable while that layer is active; a transparent binding at the corresponding position is often appropriate.

The screenshot displays the KeyBoard Editor interface. On the left is a sidebar with a menu containing: Standard Keys, Pointing Devices, Special Keys, Layer Keys, Mouse Keys, Tap Dance Keys, Macro Keys, Alt-Repeat, Leaders, Combos, Overrides, Layouts, Trainer, Quick Start, About, and Settings. The main area shows two keyboard layouts. The top layout, 'Layer 0', is highlighted with a green circle and features green keys with standard QWERTY and function keys. The bottom layout, 'Layer 1', is highlighted with an orange circle and features orange keys with various function and navigation keys. At the top of the main area is a 'Connect' button and a row of icons for different input devices. Below the keyboard layouts are buttons for 'NORMAL', 'MEDIUM', 'SMALL', 'AUTO', and 'AUTO' (with a keyboard icon).

Build a simple navigation layer

1. Choose an unused layer, such as layer 1, and assign a small group of arrow keys from Standard Keys. Make the other positions transparent if they should inherit the base.
2. Return to the base layer and select a spare position for entering navigation.
3. Open **Layer Keys**, choose **Momentary (MO)**, and assign the tile for layer 1 to that position.
4. Review and apply the edits, then deselect the target. In a text editor, hold the new key and try the arrows. Release it and confirm normal typing returns.

For a tap action on the same key, assign its desired tap key first, keep that position selected, then choose **Layer Tap (LT)** and its destination. The composer derives the tap action from the selected position.

Layer Keys choice	When to use it
Momentary · MO	Activate a layer only while held.
Layer Tap · LT	Send a key on tap; activate a layer on hold.
Toggle Layer · TG	Turn a layer on or off with a tap. Keep a way to toggle it off.
Tap Toggle · TT	Hold momentarily, or tap the configured number of times to toggle.
One Shot Layer · OSL	Activate a layer for the next key.
To Layer · TO	Replace other active layers while retaining the default. Plan a return binding.
Default Layer · DF	Change the base for this power session.
Persistent Default Layer · PDF	Save the chosen base across power cycles.

Name, inspect, and reuse layers

Click a layer name to rename it. Use its color control to make related layers easier to recognize. The layer's actions menu offers **Copy Layer**, **Paste Layer**, **Make All Blank**, **Make All Transparent**, and conversions between blank and transparent keys. Whole-layer operations affect many positions; check the target layer first.

Show Multiple Layers helps compare layers. **3D View** changes their presentation. **Hide All Transparent Keys** helps inspect inherited bindings; it does not rewrite those transparent keys. **Show All Layers** reveals layers hidden by the transparent-layer filter.

1. Open a layer's actions menu and choose **Save Layer...** to put a reusable copy in the local **Layouts** library.
2. Open **Layouts** and expand a saved or imported layout.
3. Drag the desired layer onto the destination layer in the editor.
4. Read the replacement confirmation and confirm only when the destination is correct.

5. Review and apply the resulting changes according to your update mode.

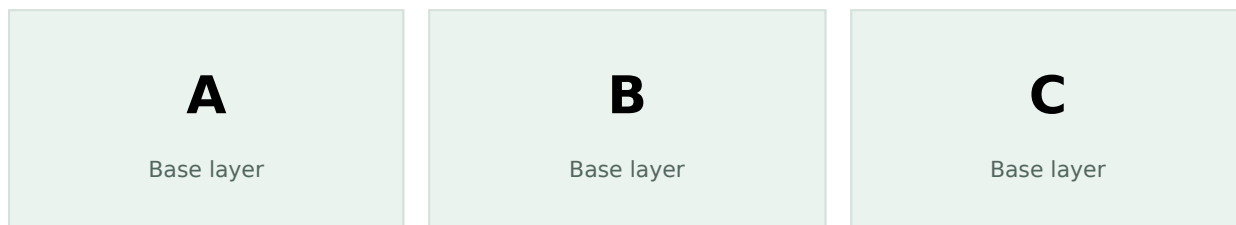
The screenshot displays the 'keyboard' application interface. On the left, a sidebar lists various configuration categories, with 'Layouts' currently selected. The main area is divided into two panes. The left pane, titled 'Layouts', shows a list of layers under the 'sval-default' group. It includes '0 Layer 0' (green), '1 Layer 1' (orange), '2 Layer 2' (blue), and '15 Layer 15' (pink). Each layer is accompanied by a small keyboard diagram showing the key assignments for that specific layer. The right pane provides a detailed view of 'Layer 0', showing a full keyboard layout with keys colored green to indicate their assignment to this layer. At the top of the right pane, there is a 'Connect' button and a row of layer numbers (0-12). At the bottom of the right pane, there are buttons for 'NORMAL', 'MEDIUM', 'SMALL', 'AUTO', and a toggle for 'AUTO'. The interface is clean and modern, with a light gray background and distinct colors for different layers.

The Layouts library holds reusable layers. Its Import control adds library content; it is separate from importing a configuration onto a board.

Local library, not a backup service. Saved layers and imported library items belong to this browser's local storage. Export important configurations to files; clearing browser data can remove local library content.

Try it: transparent is different from blank

A schematic of three positions, not a live keyboard. The base is A · B · C. Layer 1 is 1 · Transparent · Blank.



With only the base active, these three positions send A, B and C. Activate Layer 1 to see what changes.

CHAPTER 05 / BUILD USEFUL BEHAVIORS

Give a key more than one job

Special Keys groups additional actions such as media and system controls; **Layer Keys** and **Mouse Keys** have their own palettes. Use the same select-and-assign or drag-and-drop interaction as Standard Keys.

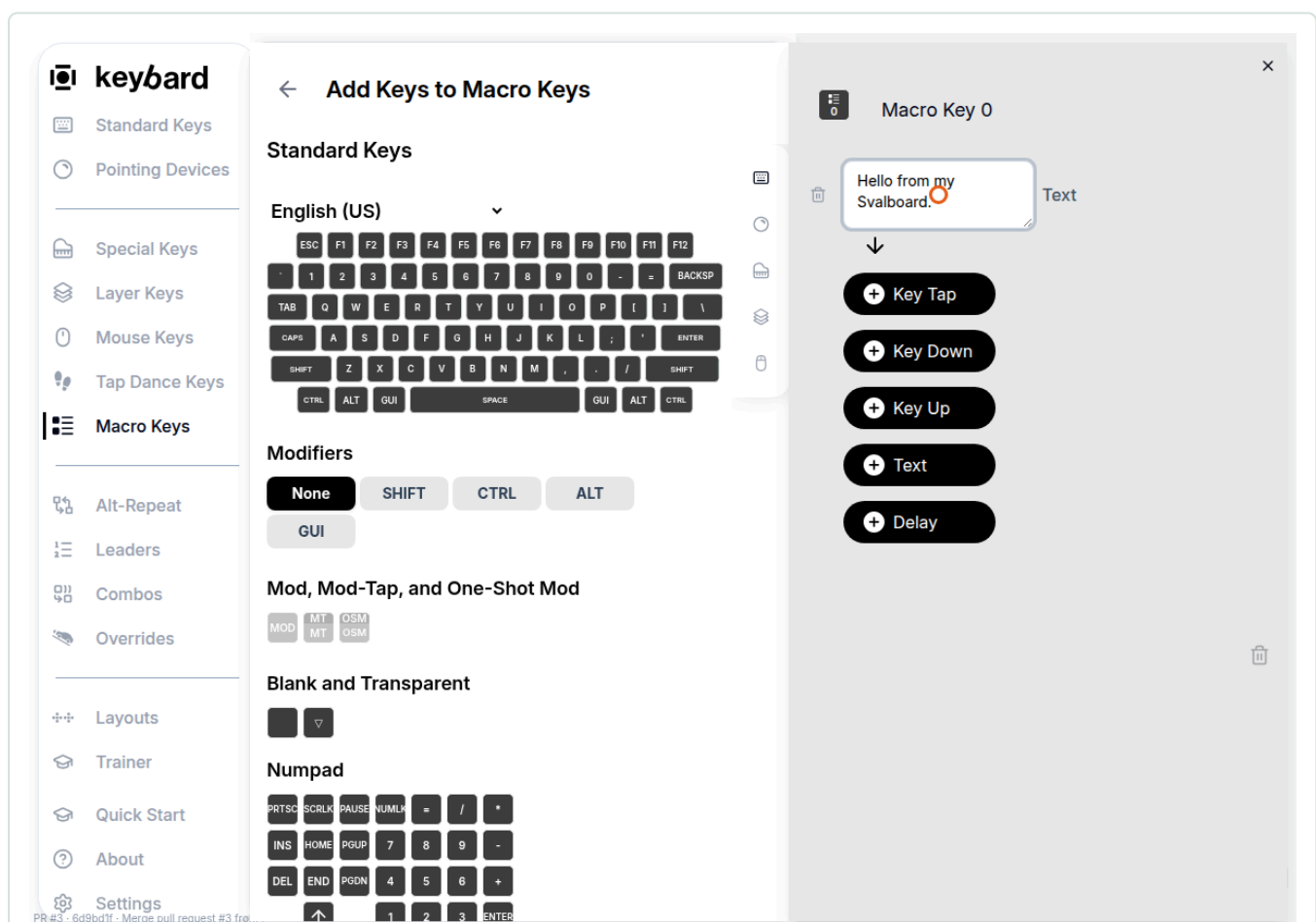
A normal assignment sends a key. A behavior can send a sequence, distinguish a tap from a hold, or recognize several keys together. Start with one small example, verify it in an empty document, and then build on it. The maintained Svalboard firmware provides 256 slots each for macros, tap dances, combos, overrides, alternate-repeat mappings, and leader sequences. Available entries depend on the firmware or file you have loaded.

Editing and using are different steps. A macro or tap dance needs both a definition and a key that invokes it. Combos and overrides respond to their configured triggers. Leaders need a Leader key as well as a sequence. In **Manual Updates**, apply the queue with **Apply N Changes** before testing on the board. In Live Updating, supported edits reach the board as you make them. Close the binding editor, deselect the assignment target, and focus a disposable text document before testing: typing while an assignment target is selected may change the layout.

Macros: send a short sequence

Macro Keys → edit a slot

A macro runs its actions in order. Begin with something harmless, such as the text “hello”. You can author a macro in an offline layout; executing it requires assigning it on a connected board.



Create a text macro. Open Macro Keys, choose slot 0, add a Text action, and enter “Hello from my Svalboard.” This creates the action; assign Macro 0 to a layout position to use it. The orange ring marks the pointer.

1. Open **Macro Keys** and edit an unused numbered entry. In the compact bottom-panel presentation, the plus control opens an empty entry.
2. Add **Text** and enter `hello` . Give the macro a useful name by clicking its title where renaming is available.
3. Close the editor. Select a spare position in your layout, then choose that macro's numbered key tile, or drag its tile onto the position.
4. Apply queued changes if necessary. Deselect the assignment and focus an empty document. Press the physical key once; it should type your text.
5. Return to the editor to add actions only after the small example works. Export a fresh backup when you are happy with it.

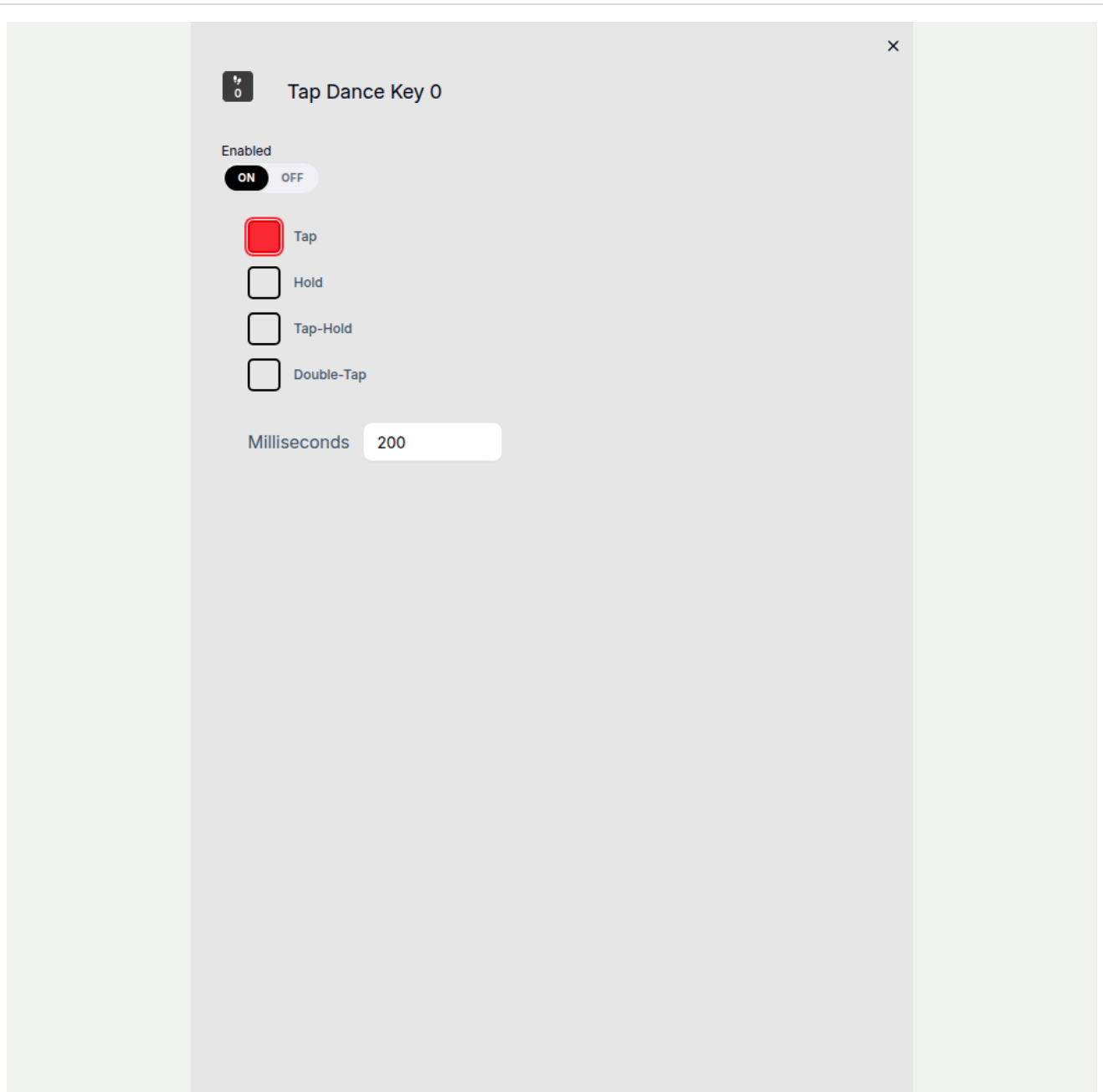
Key Tap presses and releases a key. **Key Down** holds it, and **Key Up** releases it. **Delay** inserts a pause in milliseconds. For a chord, use a modifier down, a key tap, and a matching modifier up. Always balance held keys with releases. Macro storage is shared across entries, rather than a separate fixed allowance for each macro.

Text macros send keyboard input, not clipboard contents. The operating system's keyboard layout affects the resulting characters; do not assume arbitrary Unicode text will work. Start with simple text and test punctuation in your actual OS layout. Avoid using passwords or other secrets as examples: a macro and an exported configuration are not a secure secret store.

Tap dances: tap, hold, or tap twice

Tap Dance Keys → edit a slot

A tap dance gives one physical position several outputs. A simple learning example is **Tap: A** and **Double-Tap: B**, with unused actions left empty. This makes the difference easy to see in a scratch document.



Each action has its own assignment slot. The timing field is called Milliseconds in the side panel and Hold Time (ms) in the bottom panel.

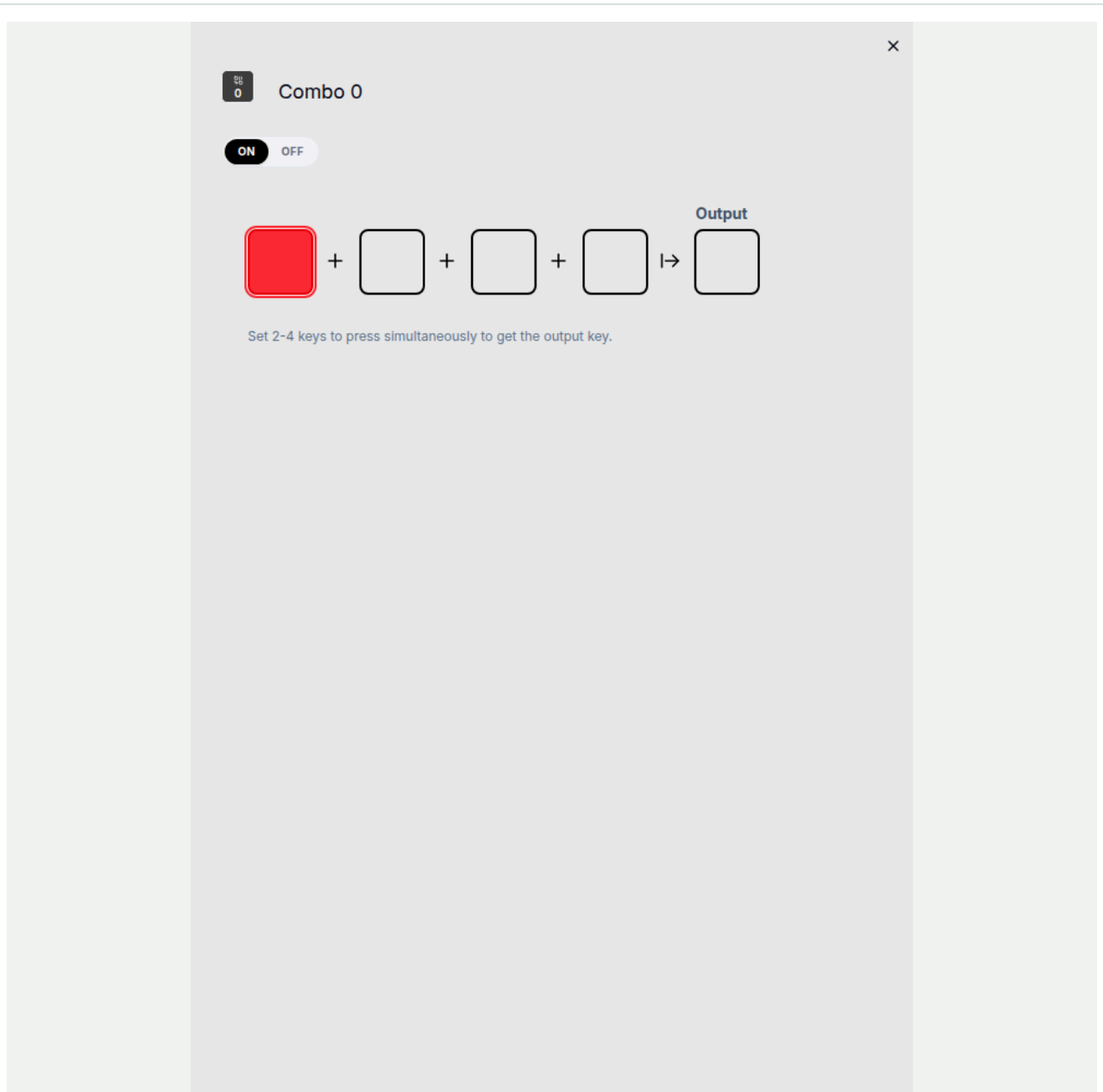
1. Open an unused entry in **Tap Dance Keys**.
2. Select the **Tap** slot and assign A. Select **Double-Tap** and assign B. **Hold** is a sustained press; **Tap-Hold** is a tap followed by a held press.
3. Review the timing field and leave **Enabled** on. Avoid changing timing and every action at once.
4. Close the editor and assign the entry's tap-dance tile to a spare keyboard position. Apply queued changes.
5. Deselect and test a single tap, then a quick double tap in an empty document. Adjust the timing if your deliberate double taps are being separated.

Release before editing. Release a held tap-dance key before changing its action. The launch firmware has a known edge case where editing a held dance can leave its previous output held.

Combos: press a small chord

Combos → **edit a slot**

A combo recognizes up to four input keycodes pressed within its timing window, and emits an **Output**. For example, A and B together can send Escape. These are keycode triggers, not a named pair of physical positions: consider where those assignments appear in your layers.



Define the inputs and output. A combo does not need a special combo key placed in the layout.

1. Choose an unused combo. Fill the first two input slots with A and B and choose Escape for **Output**.
2. Leave the remaining input slots empty and enable the entry.
3. Apply changes. Deselect, focus a safe test window, and try each key separately before pressing them together.
4. If the chord is difficult to catch, review **Combo Timeout** in the Combos panel while connected. Increase it cautiously: a generous window may catch ordinary typing rolls.

Choose combinations that are comfortable but uncommon in your normal text. After verifying the mechanics, replace the demonstration with your preferred inputs. There is no separate assignment step beyond having those input keycodes available on the keyboard.

Modifiers, mod-taps, and one-shots

Standard Keys → Modifiers → Mod, Mod-Tap, and One-Shot Mod

A modifier chord, a held modifier, and a mod-tap are different tools. A chord such as Ctrl+C sends a shortcut. A mod-tap can send A when tapped and hold Control when held. A one-shot modifier lets you tap Shift, release it, and then type a shifted key without holding both keys together.

1. For the A/Control example, first assign A to the target position and keep that position selected.
2. In **Standard Keys**, select Control under **Modifiers**. The constructed keys use the currently selected position's base assignment.
3. Choose the generated **Mod-Tap** tile. Do not choose the standalone modifier or one-shot tile by mistake.
4. Apply changes, deselect, and test a tap in a scratch document. Then hold the key while pressing another key to verify the modifier behavior.

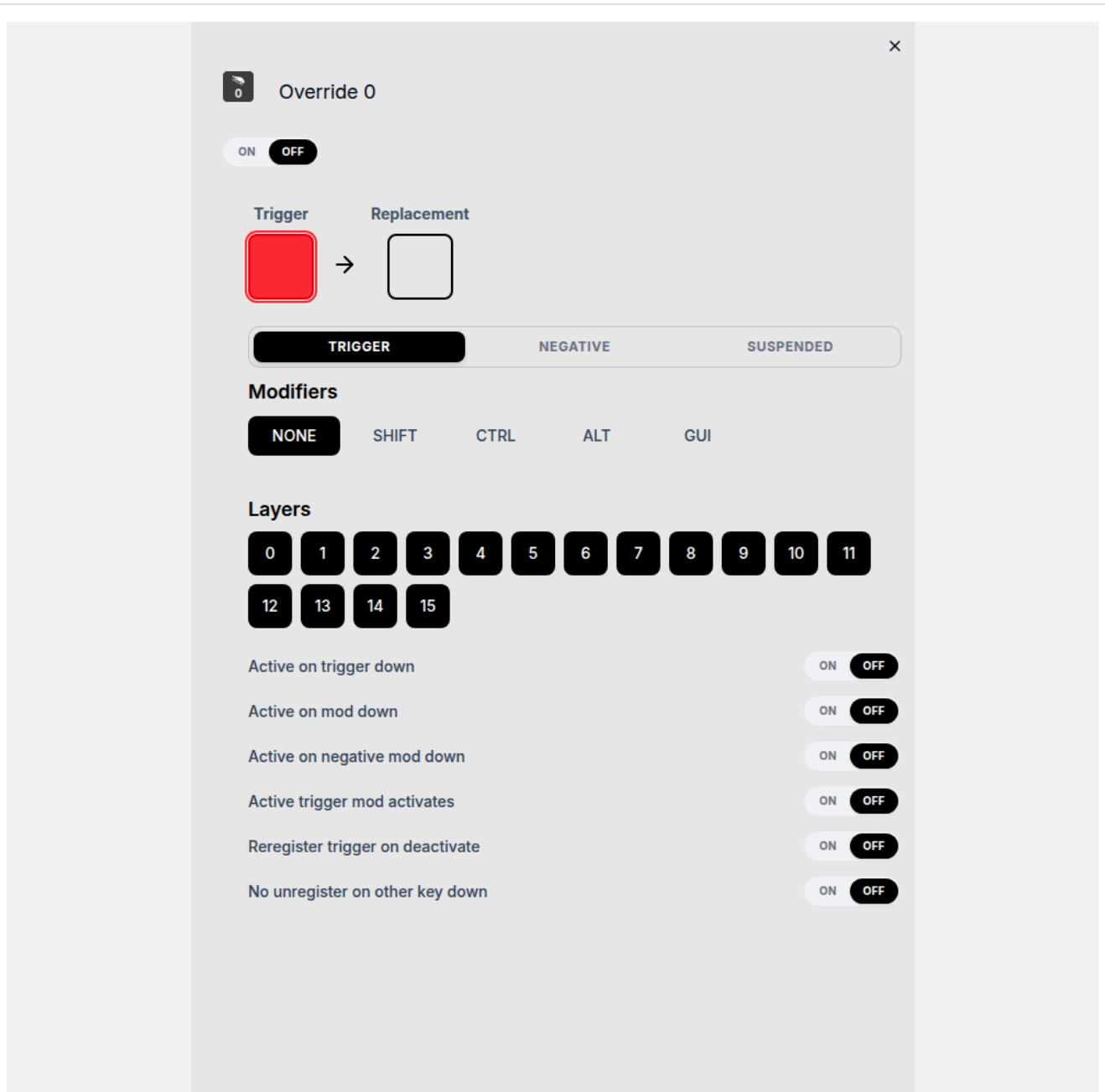
For a one-shot Shift, choose Shift and the generated one-shot tile instead. Test by tapping that key, releasing it, and typing a letter. Modifier controls include left/right choices; other composer presentations offer **MEH** (Control, Shift, Alt) and **HYPER** (those plus GUI). GUI is the Windows/Command-style modifier interpreted by your OS.

Use **QMK Settings → One Shot Keys** to review timeout and repeated-tap locking. Mod-tap decisions also depend on the Tap-Hold settings described below. Test typing rolls as well as isolated presses; a configuration that works slowly can feel different at speed.

Overrides: change an action under a modifier

Overrides → edit a slot

An override replaces a trigger when its modifier and layer conditions match. A useful example is Shift+Backspace producing Delete. It changes how an existing assignment behaves, so you do not place an “override key” in your layout.



Keep the trigger key distinct from the Trigger modifier tab: one selects the key; the other selects the required modifiers.

1. Set **Trigger** to Backspace and **Replacement** to Delete.
2. In the **Trigger** modifier tab, select the Shift modifier you use. Review handedness if you use both Shift keys.
3. Use **Suspended** for the triggering modifier you do not want included with the replacement. **Negative** specifies modifiers that prevent the override; leave it empty for this basic example.
4. Select the intended **Layers**, enable the entry, and keep advanced activation options at their existing defaults unless you have a reason to change them.
5. Apply, deselect, and test against expendable text: Backspace should retain its normal action, while the configured Shift combination should perform Delete.

Leaders: type an ordered sequence

Leaders → Leader key and sequence entries

A leader sequence is ordered input after a dedicated Leader key. Unlike a combo, its keys are typed in succession. For a harmless first example, Leader, A, B can trigger a macro that types “hello”.

0

Leader Sequence 0

ON

OFF

Sequence (up to 5 keys)

1

→

Output

Press the Leader key, then type this sequence to trigger the output keycode.

A working leader setup needs both an assigned Leader key and an enabled sequence.

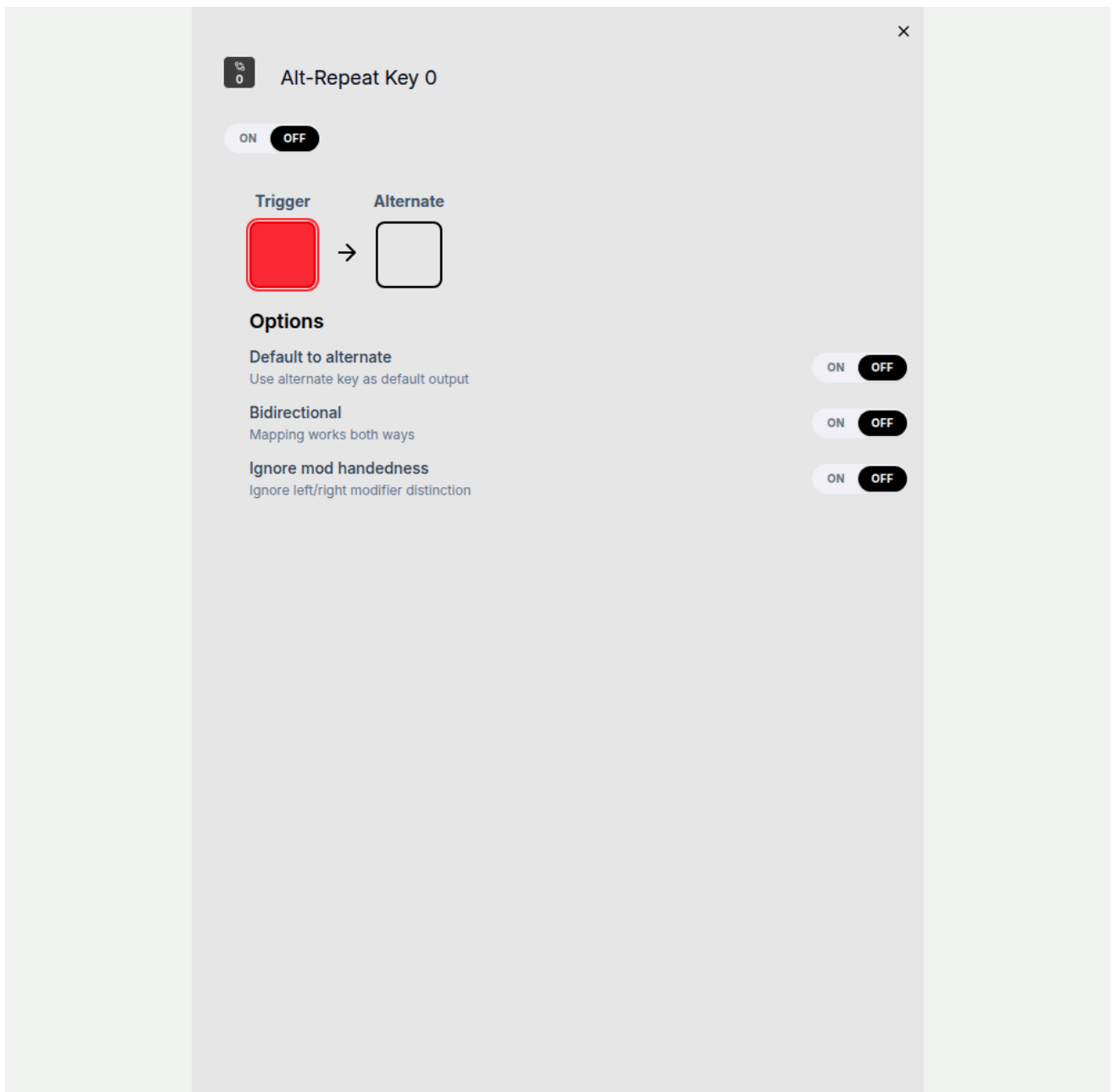
1. Create and test the simple text macro first.
2. Open **Leaders**. Assign its **Leader** tile to a spare position.

3. Edit an unused sequence. Set its first two keys to A and B, and its **Output** to your macro. Enable it. Sequences allow up to five keys.
4. While connected, review **Leader Timeout**. **Per-key timing** resets that timeout after each key; otherwise the sequence shares its time allowance.
5. Apply, deselect, focus a scratch document, and press Leader followed by A then B within the allowance.

Alternate repeat: associate two actions

Alt-Repeat → edit a mapping

Alternate repeat uses the remembered key to choose an alternate action. For exploration, map A to B and assign the alternate-repeat action to a spare position. After typing A, invoke alternate repeat and inspect the result in a scratch document.



The mapping and the key that invokes alternate repeat are separate parts of the setup.

1. Choose an unused entry, set **Trigger** to A and **Alternate** to B, and enable it.
2. Assign the alternate-repeat action from the panel to a physical position. Apply the changes and deselect.
3. Test the simple mapping before exploring **Bidirectional**, **Ignore mod handedness**, or **Default to alternate**.

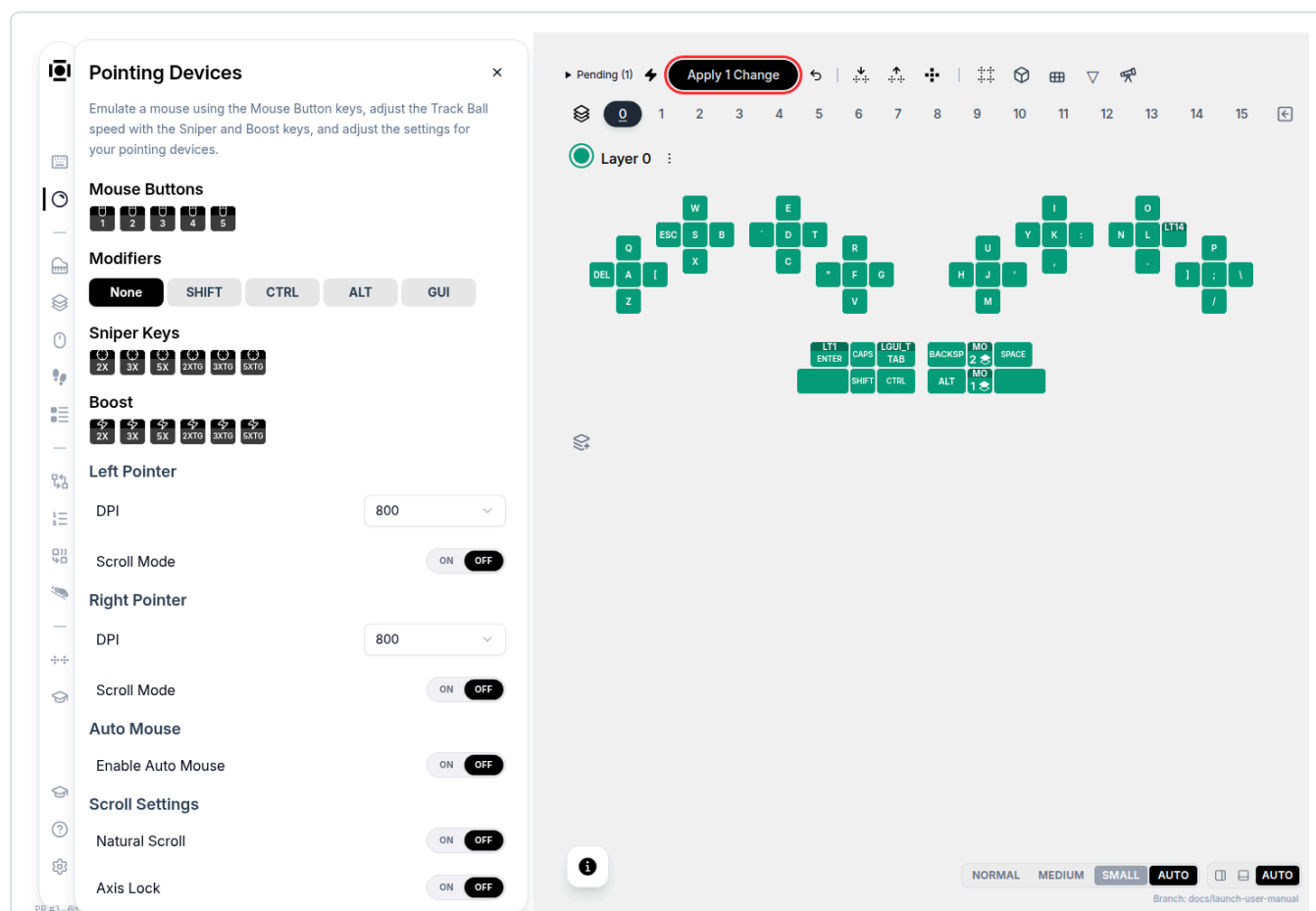
The launch firmware has known defects involving modifier matching and default-alternate behavior. Treat custom mappings as something to verify, especially with modifiers; do not depend on an untested mapping for a critical shortcut.

CHAPTER 06 / POINTING & MOUSE KEYS

Set up your pointing devices

Pointing Devices

Connect the board to see its sensor-specific controls. A loaded file can supply assignable pointing keys through the picker, but offline previews cannot demonstrate sensor behavior. The controls and ranges depend on the installed pointer, so your panel may differ from the example.



Actual connected UI with controlled test-board data. These illustrative values are not recommendations for your sensors.

1. Start with ordinary pointer movement. Adjust the left or right sensitivity in small steps, applying queued edits as needed.
2. Test scrolling and its direction. Review per-side scrolling controls, axis lock, and natural scrolling where available.
3. Assign a **Sniper** hold key to try slower, precise movement. **Boost** does the opposite. The 2×, 3×, and 5× versions choose the factor; toggle versions stay active without holding the key.
4. If you use automouse, confirm its target layer contains useful mouse bindings and a comfortable way back to ordinary typing. Check which pointer may activate it, then tune threshold, decay, and timeout one at a time.
5. Verify normal movement, scrolling, and layer return before exporting a backup.

Mouse Keys are keyboard-generated mouse movement and button actions; they are not the same as physical sensor sensitivity. TrackPoint recalibration applies to TrackPoint hardware. Avoid copying a sensitivity number between different sensor types and expecting identical movement.

Known firmware edge cases include buffered scrolling, combinations of speed factors, and disabling an active automouse layer. Keep the first configuration simple and verify the behavior you actually use.

CHAPTER 07 / SETTINGS & DIAGNOSTICS

Settings, timing, and board identity

[Settings](#) → [General](#) → [QMK Settings...](#)



QMK Settings

×

QMK Settings

Configure firmware behavior. Changes follow your update mode.

Magic

▼

Grave Escape

▼

Tap-Hold

▲

Tapling Term

Range: 0 - 10000

210

Permissive Hold (new)

☒

Hold On Other Key Press

☐

Retro Tapping (new)

☐

Quick Tap Term

Range: 0 - 10000

200

Tap Code Delay

Range: 0 - 1000

0

Tap Hold Caps Delay

Range: 0 - 1000

80

Tapping Toggle

Range: 0 - 100

5

Chordal Hold

☐

Flow Tap

Range: 0 - 10000

0

Auto Shift

▼

Combo

▼

One Shot Keys

▼

Mouse keys

▼

Leader Key

▲

▶ Pending (1)

Apply 1 Change

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SPACE

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NORMAL

MEDIUM

SMALL

AUTO

⏺

⏺

AUTO

Branch: docs/launch-user-manual

The QMK Settings panel with controlled test-board data. Check the values and supported controls reported by your own firmware.

The screenshot displays the Keyboard application interface. On the left is a sidebar menu with options: Standard Keys, Pointing Devices, Special Keys, Layer Keys, Mouse Keys, Tap Dance Keys, Macro Keys, Alt-Repeat, Leaders, Combos, Overrides, Layouts, and Trainer. The main window is titled 'Settings' and contains three tabs: General (selected), Fragments, and Developer. Under the General tab, there are sections for Appearance (with a 'Light' theme dropdown), Typing Binds a Key (with an 'ON' toggle), Serial Assignment (set to 'None'), International Keyboards (set to 'English (US)'), QMK Settings..., and Print Layers... (with a note: 'Print all non-empty keyboard layers to PDF'). On the right side of the interface is a large area showing a keyboard layout visualization for 'Layer 0'. The layout includes keys like W, E, I, O, Q, ESC, S, B, D, T, C, R, U, Y, K, N, L, P, DEL, A, [, X, Z, V, M, J, H, G, F, *, and various modifier keys like LT1, ENTER, CAPS, LGU/LT, TAB, BACKSP, MO 2, SPACE, SHIFT, CTRL, ALT, and MO 1. At the bottom right of the layout area are buttons for 'NORMAL', 'MEDIUM', 'SMALL', 'AUTO', and 'AUTO'.

Presentation preferences belong to the browser; supported keyboard settings belong to the board.

Browser choices include appearance, assignment behavior, and international palette selection. Choose a palette matching your OS keyboard layout; this does not change the OS layout itself. **Settings → Fragments** selects the finger and thumb assemblies used to draw your board. It describes the installed arrangement rather than rewiring it.

For timing, export a baseline first. Change one control, apply it, and test both slow deliberate presses and normal typing. **Tapping Term** influences tap-versus-hold decisions. **Quick Tap Term** affects repeated dual-role tapping. **Permissive Hold (new)**, **Hold On Other Key Press**, and **Retro Tapping (new)** control how other presses and releases affect that decision. **Chordal Hold** uses the hand relationship between keys; **Flow Tap** can favor taps during rapid typing and is off at zero.

Tap Code Delay controls generated tap duration, **Tap Hold Caps Delay** handles generated Caps Lock taps, and **Tapping Toggle** controls TT’s tap count. Their launch defaults are 0 ms, 80 ms, and five taps; a stored toggle count of zero behaves as one. **Grave Escape** provides separate “Always send Escape” options for Alt, Control, GUI, and Shift.

A visible setting is not a guarantee that a firmware feature is enabled. Standard launch builds do not include Auto Shift, and some mouse-key settings have no runtime consumer. Use the launch notes for those limits rather than repeatedly adjusting an ineffective field.

Where supported, **Board name** changes how the keyboard appears to your computer. **Save now** writes immediately, even in manual mode; restart the keyboard for the USB name to change. Apply or export outstanding edits before restarting. Board identity is separate from a layout backup.

Settings → Developer → Scan Lab... is for advanced scanning, power, and firmware diagnostics. Its timing application and bootloader controls can affect hardware immediately. For a simple “does this switch register?” check, use Matrix Tester instead. Power estimates are not battery-life measurements, and longer idle intervals can delay the first response after inactivity.

Check every physical switch

Editor toolbar → Matrix Tester

1. Connect the board and open **Matrix Tester** from the toolbar.
2. Press each finger direction and each thumb key, holding briefly so the polling display can see it. Compare the highlighted position with the one you pressed.
3. Use **Clear Matrix** to clear the visual test history and try again. It does not erase your layout or board settings.
4. Exit Matrix Tester to return to editing.

This is a physical-switch check, not a monitor of the final text produced by macros, combos or tap dances. Very brief presses can fall between snapshots. If the matrix sees the switch but typing is wrong, inspect its layer and binding rather than assuming the switch failed.

Print a desk reference

Open **Settings → General → Print Layers....** Review the non-empty layers, then use the browser’s print dialog to print them or save a PDF. A printed diagram is a learning reference; export a `.svil` as well for an editable backup.

Back up and import with confidence

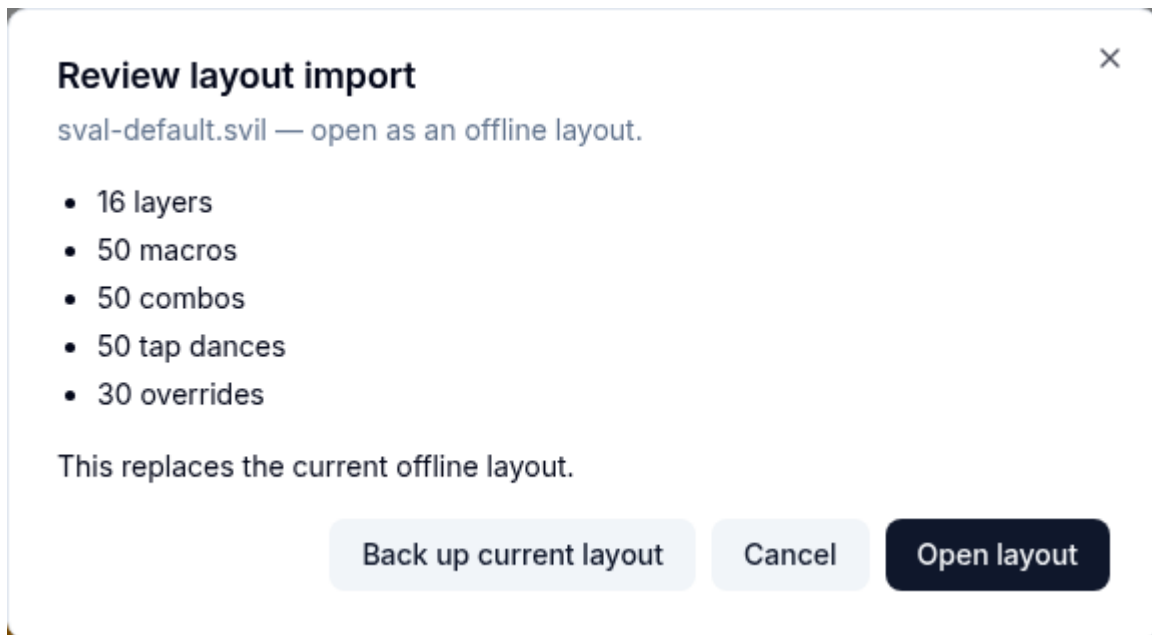
Editor toolbar → Export Layout or Import Layout

Use native `.svil` for your working backups. Keep **Include Macros** enabled unless you intentionally want to omit macro content. The legacy `.vil` export retains QMK settings, but does not preserve all Sval-specific hardware values and metadata, including names, colors, and cluster selections.

Operation	What it preserves or changes
Export Layout	Downloads the editor's current configuration, including staged work.
Load File on welcome screen	Opens an offline draft; does not write a board.
Import Layout in editor toolbar	Reviews configuration changes for the current target.
Import in Layouts panel	Adds reusable layouts to the browser library.
Save Layer...	Saves one layer locally for reuse, not a complete device backup.

Import a compatible backup onto a connected keyboard

1. Connect the destination board and export its current configuration before making changes.
2. Apply or discard pending edits before beginning an import.
3. Choose **Import Layout**, select the file, and read **Review layout import**.
4. Check the destination, counts, warnings, and errors. **Back up current layout** is available in the review.
5. In Manual mode, choose **Stage import**, review pending changes, then **Apply**. In Live Updating, **Apply import** writes the changes. Offline, **Open layout** replaces the current draft.
6. Check the imported bindings and test important actions before relying on them.



An offline import review. A connected review identifies the keyboard destination and offers Stage import or Apply import according to the update mode.

Import is not a complete device-image restore. The current importer keeps the board's alt-repeat entries, leaders, one-shot configuration, layer colors, and fragment state. It reports unsupported fields and settings. Entries beyond the incoming file's layers and tables are kept rather than erased.

A file may be rejected for mismatched matrix dimensions, too many layers or table entries, incompatible keycodes, unsupported macro content, or capacity limits. Read the specific message instead of forcing the file through. Native backups preserve information even when a particular field cannot yet be restored automatically.

CHAPTER 09 / THE DESKTOP TRAINER

Trainer and the desktop overlay

Trainer, directly below Layouts

Trainer previews the appearance and controls the optional desktop overlay. The browser is useful for choosing a style or reviewing an offline layout. Keyboard Host supplies the native transparent window above your other applications and reads live board state directly. Its board operations are read-only.

keyboard

Standard Keys

Pointing Devices

Special Keys

Layer Keys

Mouse Keys

Tap Dance Keys

Macro Keys

Alt-Repeat

Leaders

Combos

Overrides

Layers

Trainer

Quick Start

About

Settings

Trainer

Use Trainer on your desktop

Install Keyboard Host to show the overlay above your other apps.

Download for Windows

Extract the ZIP, then run **Start-Windows.cmd**. First launch needs Internet to install its local runtime. No administrator access needed.

The app opens Keyboard in your browser. Connect your board, then select **Trainer**. The overlay stays running in the system tray.

Is Keyboard Host running? Connect this page to it. If your browser asks to let this site access apps on your device, allow it.

Connect to Keyboard Host

Already running? Open local Keyboard

Release notes

Preview background

LightDarkBusy

Layout source

Loaded snapshot

Import for trainer

Default layer

Preview layer

0 - Layer 0

0 - Layer 0

OverlayAppearanceFeedbackPractice

Preset

Outline only

Key fill

0%

Outline

86%

Legend

100%

Outline thickness

1 px

Contrasting legend halo

Layer change

Pressed key

PR #3 - 6d8bd1f - Merge pull request #3 from svalboard/refactor/keyboard-service-names

The browser preview, not a captured native desktop overlay. Light, Dark, and Busy test backgrounds help you judge contrast.

The browser preview, not a captured native desktop overlay. Light, Dark, and Busy test backgrounds help you judge contrast.

Keep your work in view.
Keep your layout nearby.

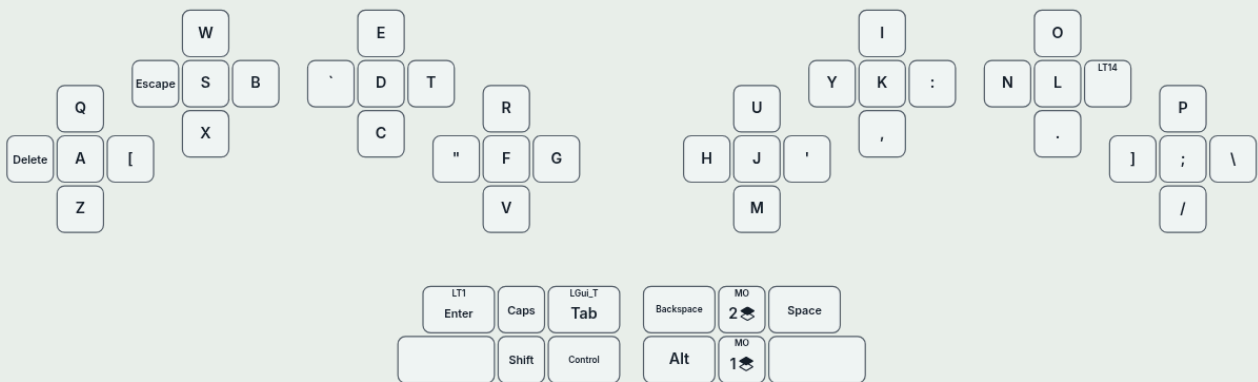


Illustration using the actual native renderer with simulated host state and the bundled example. The background is illustrative; this is not a capture of a running Windows desktop.

Install and show the Windows overlay

1. Select **Download for Windows** at the top of Trainer. Extract the ZIP and run **Start-Windows.cmd**.
2. Allow the first launch to download its private runtime. It needs internet, but no administrator rights, system Python, or PATH changes.
3. The app opens Keyboard. Connect your keyboard normally and select Trainer. If needed, use **Connect to Keyboard Host** and allow the browser's local-device access request. The install panel also links to local Keyboard when Host is running.
4. Choose the board in the host selector and use **Show overlay**. Host remembers its selected board for reconnection.
5. Drag the overlay into place. In **Overlay**, optionally enable **Click through keyboard**; the small handle remains available for moving or hiding it. **Place at bottom** provides a convenient starting position.

The companion stays in the system tray when you close the browser. Use **Hide overlay** or the overlay's handle controls to get it out of the way. After editing the keyboard layout, choose **Reload layout** so the companion refreshes its cached copy. If reconnection fails, check the host board selection and USB connection rather than importing a different layout to mask the problem.

The Windows download is a portable preview, not a signed installer. Linux source launch is available, with Wayland limitations for placement, stacking, click-through, and modifier reporting. macOS remains unverified; equivalent validated packages for all platforms are not promised.

Choose what you can see

Start with an **Appearance** preset, then adjust **Key fill**, **Outline**, and **Legend** colors and opacity independently. Try **Contrasting legend halo** over busy backgrounds. Use **Outline thickness** for definition without making the fill more opaque. **Hands** and **Overlay size** are in Overlay.

In **Feedback**, choose **Layer-change highlight**: Off, Quick flash, or Short fade, and its duration. Enable **Highlight held keys** for a matrix-tester-like indication. Matrix polling can miss very brief taps and does not describe every generated macro or combo output. The preview's **Preview held keys** button simulates a chord.

Current firmware reports active and default layers automatically, including changes during use. Older firmware may require **Default layer (older firmware)**. For offline previews, choose **Default layer** and **Preview layer**. **Layout source** distinguishes a loaded snapshot, editor draft, example, imported trainer layout, and a live read-only board. **Import for trainer** changes the preview source; it does not write that file to the keyboard.

Practice recall without recording your typing

The screenshot shows the 'Trainer' section of the 'keyboard' application. On the left is a sidebar menu with categories like 'Standard Keys', 'Pointing Devices', 'Special Keys', 'Layer Keys', 'Mouse Keys', 'Tap Dance Keys', 'Macro Keys', 'Alt-Repeat', 'Leaders', 'Combos', 'Overrides', 'Layouts', 'Trainer' (selected), 'Quick Start', 'About', and 'Settings'. The main content area has a header 'Use Trainer on your desktop' with instructions and a 'Download for Windows' button. Below this is a 'Connect to Keyboard Host' button and links for 'Already running?', 'Open local Keyboard', and 'Release notes'. The central part features a large keyboard layout preview with a 'Preview background' section (Light, Dark, Busy) and 'Layout source' (Loaded snapshot, Import for trainer). Below the preview are 'Default layer' and 'Preview layer' dropdowns, both set to '0 - Layer 0'. On the right, a 'Practice' tab is active, showing 'Recall practice' (enabled) and 'Familiar bindings' (disabled). The 'Recall practice' section describes self-assessed practice. The 'Familiar bindings' section includes a 'Select a binding' dropdown, 'Mark familiar', 'Clear', and 'Hide familiar legends' toggle.

Practice is self-assessed: reveal the location, then tell Trainer whether you remembered it.

1. Open **Practice** and enable **Recall practice**. Read the requested binding and hand, and decide where you would press.
2. Choose **Reveal**, then **Remembered** or **Again** to advance. This is not an automatic typing test.
3. For selective help, choose a key in the preview or the binding selector, click **Mark familiar**, and enable **Hide familiar legends**.
4. Turn practice off to return to ordinary reference. Closing the temporary controls also returns the desktop overlay to reference mode.

Recall counts and familiar selections are temporary practice state, not a durable learning-history service. Appearance preferences are saved separately. When following the board, recall can also hide desktop legends; otherwise you are practicing against the selected preview layout.

Return to your work

After a refresh, Keyboard intentionally returns to the welcome screen. Previously permitted keyboards can appear under **or reconnect without the chooser**. Select the correct device to read its saved state again.

To continue offline work, choose **Load File** and reopen your exported draft. Browser preferences and the local Layouts library can persist, but they are not an automatic recovery copy of the current editor session.

Before each editing session, check the target, update mode, selected layer, and whether a key is selected. Use a small naming convention for backups: one before a major change, one after verification. Keep both until the new layout is comfortable.

CHAPTER 11 / TROUBLESHOOTING

Recover from common problems

I cannot connect

Check WebHID browser support, the USB data connection, the chosen device, and compatible firmware. Close other configuration applications if they are holding the device, then reconnect. A previously granted browser permission is not proof that the cable, firmware, or current device connection is working.

The layout changed, but typing did not

Check whether you are editing an offline draft or have pending Manual changes. Confirm the physical keyboard is on the layer you edited. Transparent keys may inherit a different binding, and a layer-tap key waits to distinguish a tap from a hold.

Typing unexpectedly changed a binding

Deselect the target and check **Typing Binds a Key**. In Manual mode, inspect and discard unwanted pending edits. If a change was already applied, explicitly restore the previous binding; discarding does not undo saved writes.

A save failed

Read the **Not saved** message. Pending edits remain, and some earlier writes may already have succeeded. If still connected, use **Retry**. If the device disconnected, **export the retained**

draft before reconnecting. Reconnection reads the board again and may discard pending edits. Compare that state with your saved draft, then review an import if appropriate, observing the restore limitations above.

A name will not save

Layer, macro and tap-dance names have a 16-byte UTF-8 limit on supported firmware. Accented characters and emoji can use several bytes each. Shorten the name and read the error. A board's USB name is a separate setting with its own limits.

Controls or layers seem missing

Expand the navigation rail, reveal **Show editor controls** if the toolbar is collapsed, or use **Show All Layers**. Some settings depend on firmware capabilities and attached hardware; an offline example cannot demonstrate every connected-device function.

Keyboard · Svalboard-QMK launch manual

Browser state, files and the board are different places to keep your work. Check the editing target, make one change at a time, and keep an exported copy you have verified.

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